

Han Song

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TECH SKILLS

- Languages: Python, Java, C/C++, SQL, JavaScript, TypeScript
- Databases: MySQL, MongoDB, Neo4j, Redis
- Developed Tool kits: Apache Kafka, Apache Flink, ClickHouse, Docker, Kubernetes, RestAPIs, Elasticsearch, Falcon, Flask, Azure Cloud, Kibana, Postman, gRPC, VueJS/Vuex, React/Redux, NodeJS, NestJS, Prisma, Vuetify, Material-UI, Bootstrap, Echarts, d3.js, HubSpot, Swagger, Postmark

WORK EXPERIENCE

Software Engineer & Data Engineer, Vibrant Wellness

Dec 2020 – present

Santa Clara, CA, USA

Large-Scale Data Ingestion & Migration

- Enhanced the large-scale data **ingestion pipeline**, ensuring clean, filtered, and well-integrated data to improve accuracy and consistency.
- Built the development of order data processing system handling **40K+** active user datasets.
- Implemented an order upload service and migrated over **1 million** historical order data from the company's legacy platform to customer relationship management platform.

Real-Time Data Pipeline & Quality Assurance System

- Led a post-write pipeline with **Apache Kafka** and **Flink** for the core ingestion service, enabling asynchronous and independent workflows in post-write processing, significantly improving **real-time data streaming**.
- Incorporated **audit logging and automated notifications**, enabling seamless integration with downstream services while maintaining data reliability.
- Implemented an **automated data verification and correction service**, ensuring data consistency and accuracy using **ClickHouse** and **Airflow**. Designed a proactive error detection system that promptly notifies upstream systems for timely resolution, achieving a monthly error rate of less than 0.5% across all dimensions.

Laboratory Information Management System

- Developed a **role-based access control** service, enhancing internal security and operational efficiency by simplifying permission management for over 10 internal groups.
- Designed and implemented the customer-facing **order tracking** system with **microservice** to verify identity at different role levels and track sample kit status, ensuring scalability, security, and regulatory compliance.
- Automated the order placement workflow by integrating with FedEx **webservice API** into the order management system. This automation reduced manual tracking errors and accelerated batch order processing.

Sales Information Management System

- Collaborated with Sales and Marketing teams to lead the development of the **sales dashboard** with critical business metrics, incorporating projection and prediction capabilities to empower them to make **data-driven business strategies**.
- Designed and implemented an **internal ticket system** to optimize project management and issue tracking, significantly boosting team collaboration and operational efficiency across 8+ teams.

EDUCATION

University of Pittsburgh, Pittsburgh, PA

Aug 2018 – May 2020

Master of Science in Information Science, School of Computing and Information

Main Course: Algorithm Design, Advanced Topic in Database Management, Interactive System Design

Sichuan University, Chengdu, China

Sep 2013 – Jun 2017

Bachelor of Engineering in Software Engineering, School of Computer Science and Software

Main Course: Operating System, Software Engineering, Computer Network, Object-Oriented Programming & Design